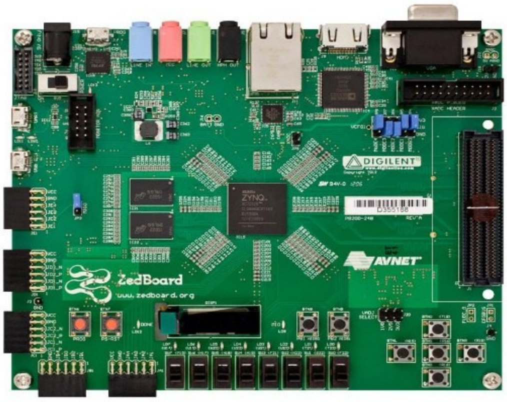
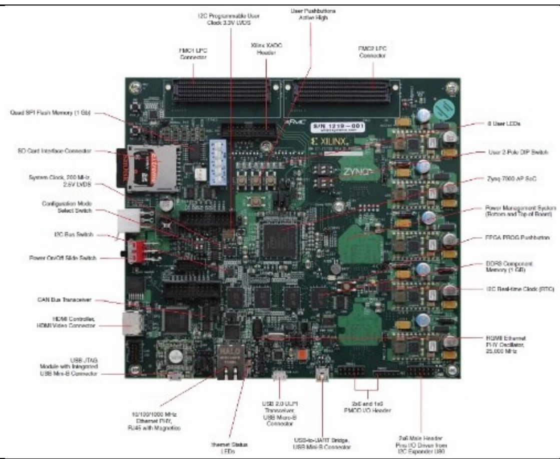
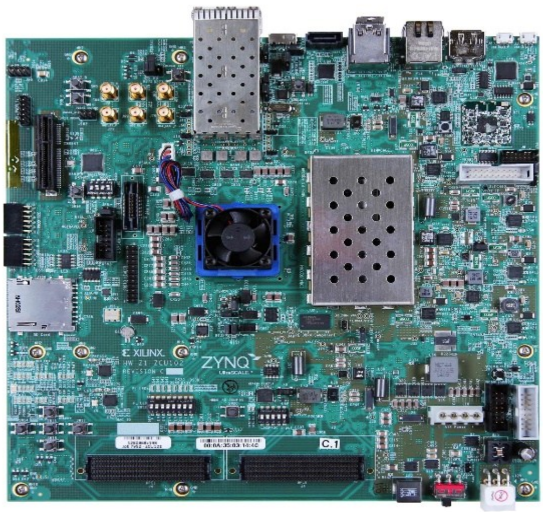
NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Electronics And Communication Engineering Department

VLSI Design Laboratory Equipment

<table>
<tr><th colspan="2">Advanced Hardware & FPGA Equipment Overview</th></tr>
<tr><td>DIGILENT ZedBoard:
The Digilent ZedBoard is a cost-effective, high-performance evaluation and development board built around the AMD/Xilinx Zynq-7000 All Programmable SoC (AP SoC). It is an engineered collaboration between Digilent, Avnet, and other industry partners. The board is widely used in academia and industry to design complete embedded hardware-software systems.</td><td></td></tr>
<tr><td>Xilinx Zynq-7000 AP SoC ZC702 Evaluation Kit:
The AMD Xilinx Zynq-7000 SoC ZC702 Evaluation Kit is an industry-standard, comprehensive development platform designed for prototyping high-performance embedded processing systems. Built directly by Xilinx (now AMD), this kit provides engineers with all the necessary hardware, pre-verified reference designs, and IP to evaluate the Zynq-7000 All Programmable SoC architecture.</td><td>
</td></tr>
<tr><td>Zynq Ultrascale+™ MPSoC ZCU102 Evaluation Kit:
The AMD Xilinx Zynq UltraScale+ MPSoC ZCU102 Evaluation Kit is a premier, enterprise-grade development platform engineered for heterogeneous Multi-Processor System-on-Chip (MPSoC) prototyping. It provides a massive leap in processing power, memory bandwidth, and high-speed connectivity over previous generations. This makes it an industry-standard choice for complex artificial intelligence, automotive, networking, and 4K multimedia designs.</td><td></td></tr>
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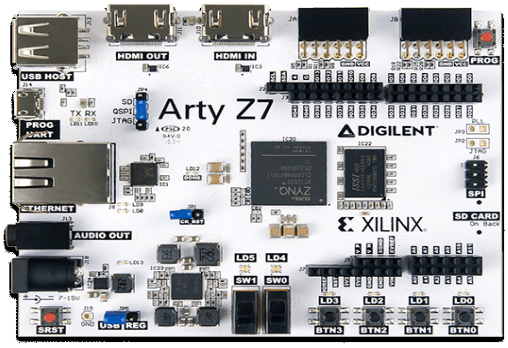
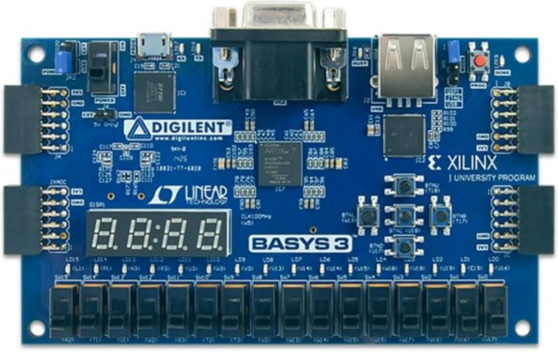
DIGILENT Arty Z7: The **Digilent Arty Z7** is a compact, maker-focused development board designed around the **AMD/Xilinx Zynq-7000 All Programmable SoC (AP SoC)**. Tailored specifically for hobbyists, educators, and rapid prototyping, it packages professional-grade processing capabilities into an accessible, budget-friendly layout featuring **Arduino and Pmod expansion options**.	
DIGILENT Basys 3: The **Digilent Basys 3** is one of the most popular entry-level FPGA trainer boards in the world. It is built exclusively around the **AMD/Xilinx Artix-7 FPGA** architecture and is specifically designed for students, educators, and beginners learning digital circuit design and Hardware Description Languages (HDL) like VHDL or Verilog.	
CPLD Development Platform: ST105 A **CPLD (Complex Programmable Logic Device) Development Platform** refers to a comprehensive hardware ecosystem and modular motherboard architecture designed for testing, verifying, and deploying non-volatile programmable logic. The **Scientech 105 CPLD Development Platform** (commonly designated as **ST105**) is an educational and prototyping hardware trainer board centered around the legacy **Xilinx XC9500 series CPLD architecture**. Manufactured primarily for technical universities, electronics labs, and vocational institutes, the platform is designed to provide an expandable environment for engineering students to design, simulate, and physically implement digital logic circuits using VHDL or Verilog.	

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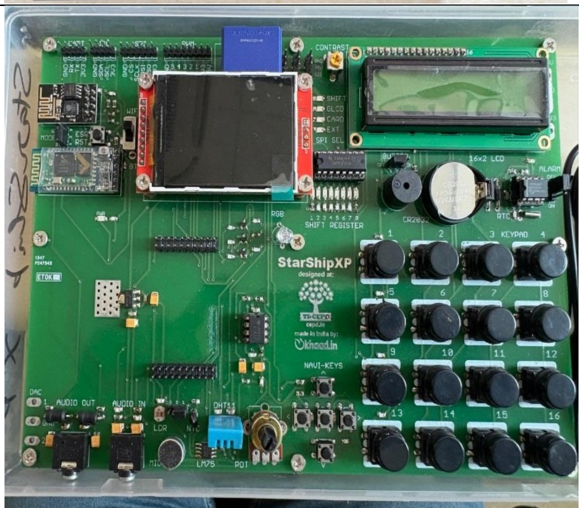
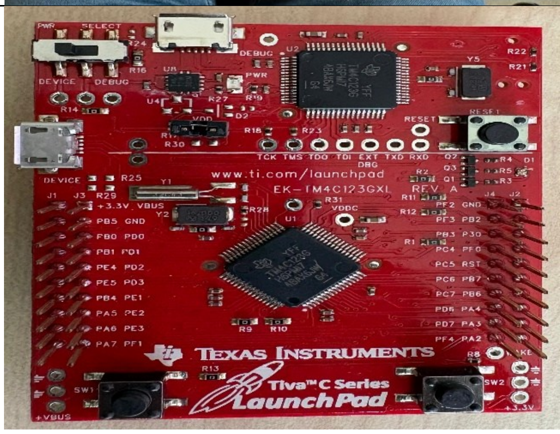
FPGA Trainer Kit: EST-FPGA An **EST FPGA** typically refers to an **Embedded System Trainer (EST) FPGA platform**, which is an all-in-one educational and industrial prototyping hardware ecosystem. Rather than being a standalone chip or a minimalist development board, an EST platform (such as the widely used *AnshumanTech XPO-EST Series* or similar VLSI engineering lab trainers) is engineered as a comprehensive testbed.	
Cross-platform Motherboard: StarShip XP A **cross-platform motherboard** is a specialized baseboard designed to accept multiple types of interchangeable processor modules or microcontrollers. A cross-platform motherboard serves as a standardized hardware testing infrastructure. It decouples the core processing silicon from the peripheral I/O subsystems, allowing engineers to benchmark, test, and swap entirely different microcontrollers.	
TIVA launchpad Evaluation Kit: EK-TM4C123GXL The **Texas Instruments EK-TM4C123GXL Tiva C Series LaunchPad** is an evaluation platform built specifically for the **TM4C123GH6PM** microcontroller. It is one of the most widely used development boards in embedded system education and prototyping due to its combination of high-speed calculation, deterministic ARM Cortex architecture, and low price point.	
Test, Measurement & Analysis Gear	

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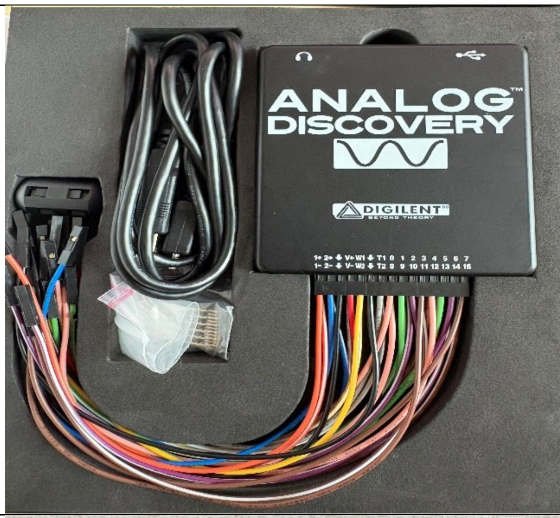
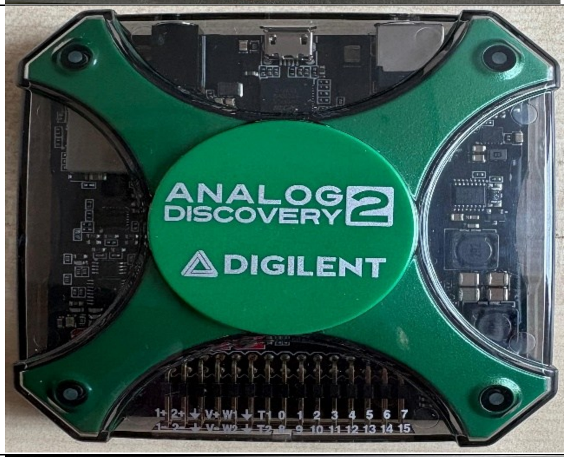
DIGILENT Analog Discovery Design Kit: The **Digilent Analog Discovery** (developed in collaboration with Analog Devices Inc.) is a pocket-sized, multi-function test and measurement device that transforms any PC into a fully featured electronics engineering laboratory. It allows students, hobbyists, and field engineers to build, stimulate, measure, and analyze mixed-signal circuits anywhere without needing a stack of bulky, expensive benchtop equipment.	
DIGILENT Analog Discovery 2 Design Kit: The **Digilent Analog Discovery 2 Design Kit** (engineered in collaboration with Analog Devices) is an all-in-one, pocket-sized USB test and measurement platform. It effectively compresses a traditional electronics engineering lab bench into a single rugged device, allowing students, hobbyists, and field engineers to stimulate, analyze, and debug mixed-signal electronic circuits from any computer.	
Logic Analyzer: KEYSIGHT 16862A The **Keysight 16862A** Portable Logic Analyzer is an industry-leading, high-performance diagnostic instrument engineered for validation, debugging, and optimization of complex high-speed digital systems. As part of the prestigious 16860 Series, it delivers the industry's fastest deep-memory timing capture in a portable form factor. It is heavily utilized by digital design engineers to analyze advanced memory buses (such as DDR/LPDDR) and embedded system buses.